

HERNANDO, PANTINE B. CHALLENGE - 19 - CRYPTOGRAPHY

BSIT-3

HideToSee 

Medium

Cryptography

picoCTF 2023

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Hints 

Description

1

How about some hide and seek heh?

Look at this image [here](#).

14,624 users solved

 43% Liked 

 picoCTF{FLAG}

Submit Flag

1. Challenge Overview

The challenge presents an image and the clue *"How about some hide and seek heh?"* The title **"HideToSee"** points toward steganography—the practice of concealing a file, message, image, or video within another file.

2. Methodology & Execution

Step 1: Extraction via picoCTF Webshell

Instead of using local tools, we utilize the **picoCTF webshell** to interact with the file in a Linux environment.

Download: Use **wget** to pull the image directly into the webshell instance.

Bash

```
wget https://artifacts.picoctf.net/c/235/atbash.jpg
```

1. **Extract:** Use **steghide**, which is pre-installed on the webshell, to extract hidden data. When prompted for a passphrase, simply press **Enter** (leave it blank).

Bash

```
steghide extract -sf atbash.jpg
```

2. **Read:** List the files to find **encrypted.txt** and display its contents.

Bash

```
Pantine23-picocftf@webshell:~$ wget https://artifacts.picocftf.net/c/241/atbash.jpg
--2026-04-05 13:35:22-- https://artifacts.picocftf.net/c/241/atbash.jpg
Resolving artifacts.picocftf.net (artifacts.picocftf.net)... 3.170.131.18, 3.170.131.77, 3.170.131.72, ...
Connecting to artifacts.picocftf.net (artifacts.picocftf.net)|3.170.131.18|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 51510 (50K) [application/octet-stream]
Saving to: 'atbash.jpg'

atbash.jpg                                     100%[=====

2026-04-05 13:35:23 (2.12 MB/s) - 'atbash.jpg' saved [51510/51510]

Pantine23-picocftf@webshell:~$ strings atbash.jpg | more
JFIF
```

cat encrypted.txt

3. **Resulting String:** **krx1XGU{zgyzhs_xizxp_9zy96555}**

```
Pantine23-picocftf@webshell:~$ file atbash.jpg
atbash.jpg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 465x455, components 3
Pantine23-picocftf@webshell:~$ steghide extract -sf atbash.jpg
Enter passphrase:
wrote extracted data to "encrypted.txt".
Pantine23-picocftf@webshell:~$ cat encrypted.txt
krx1XGU{zgyzhs_xizxp_7142uuv9}
Pantine23-picocftf@webshell:~$
```

Step 2: Decoding with CyberChef

With the encrypted string in hand, we move to **CyberChef** to handle the decryption:

1. **Input:** Paste **krx1XGU{zgyzhs_xizxp_9zy96555}** into the Input box.
2. **Recipe:** Drag the **Atbash** cipher into the recipe area.
3. **Result:** The tool reverses the alphabet, revealing the flag.

Download CyberChef

Last built: A day ago

Operations475

Atbash

Atbash Cipher

Favourites★

Data format

Encryption / Encoding

Public Key

Arithmetic / Logic

Networking

Language

Utils

Date / Time

Extractors

Compression

Hashing

Code tidy

Recipe

Atbash Cipher

Input

krx1XGU{zgyzhs_xizxp_7142uuv9}

Output

picoCTF{atbash_crack_7142fde9}

Final Flag: picoCTF{atbash_crack_9ba96555}

3. Reflection

Using the **picoCTF webshell** streamlines the process significantly, as it removes the need to install dependencies like **steghide** on a personal machine. This challenge was a great reminder that "hidden" doesn't always mean "encrypted"—sometimes the data is just tucked away in a place you aren't looking yet.

Combining the command-line extraction with **CyberChef's** visual decoding interface makes for a very efficient workflow. It's also a clever touch by the authors to name the extracted file/image **atbash**, effectively giving you the key to the final puzzle if you're paying close attention to the filenames.